Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.TECH. CE and IIIrd
Name of the Paper: Solids Mechanics
Paper Code: TCE-303
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.
(iii) Please specify COs against each question.

Q1. (10 Marks)

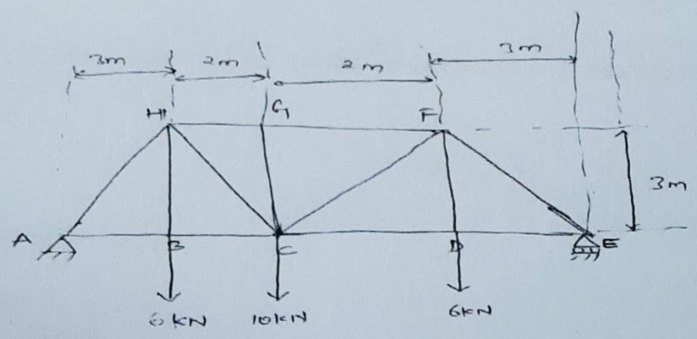
a. What is the composition and resolution of forces? Illustrate with a diagram. (CO1)

OR

b. Three forces of 50 N, 60 N, and 80 N act concurrently at a point with angles of 30°, 120°, and 240°, respectively. Find the resultant force. (CO1)

Q2. (10 Marks)

a. Determine the forces in the members BC, HC, and HG for the truss shown in Fig. State whether the forces are in tension or compression



(CO2)

OR

b. b. Explain the stress-strain behavior of steel under uniaxial loading. (CO2)

Q3. (10 Marks)

a. How are thermal stresses developed in a material, and what factors influence their magnitude? (CO2)

OR

b. Define the following term:

- Strain Energy
- Hoop Stress
- Stress

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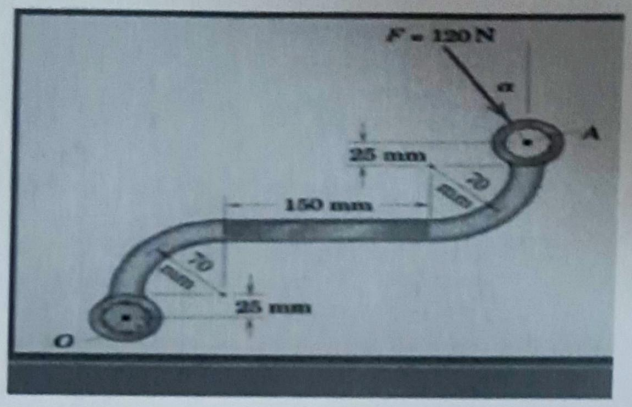
- Tension Force (CO2)

Q4. (10 Marks)

a. The 120 N force is applied as shown to one end of the curved wrench. If alpha is equal to 30 degree. Calculate the moment of F about center O of the bolt. Determine the value of alpha which would maximize the moment about O, calculate the value of this maximum moment.



(CO1)

OR

b. Derive the relation for determining the resultant and direction of forces using parallelogram.

(CO1)

Q5. (10 Marks)

a. Write a short note on the following:

- Resultant
- Equilibrium
- Free Body diagram
- Principle of Superposition (CO1)

OR

b. How does the method of joints help in analyzing trusses? Provide a detailed explanation. (CO2)